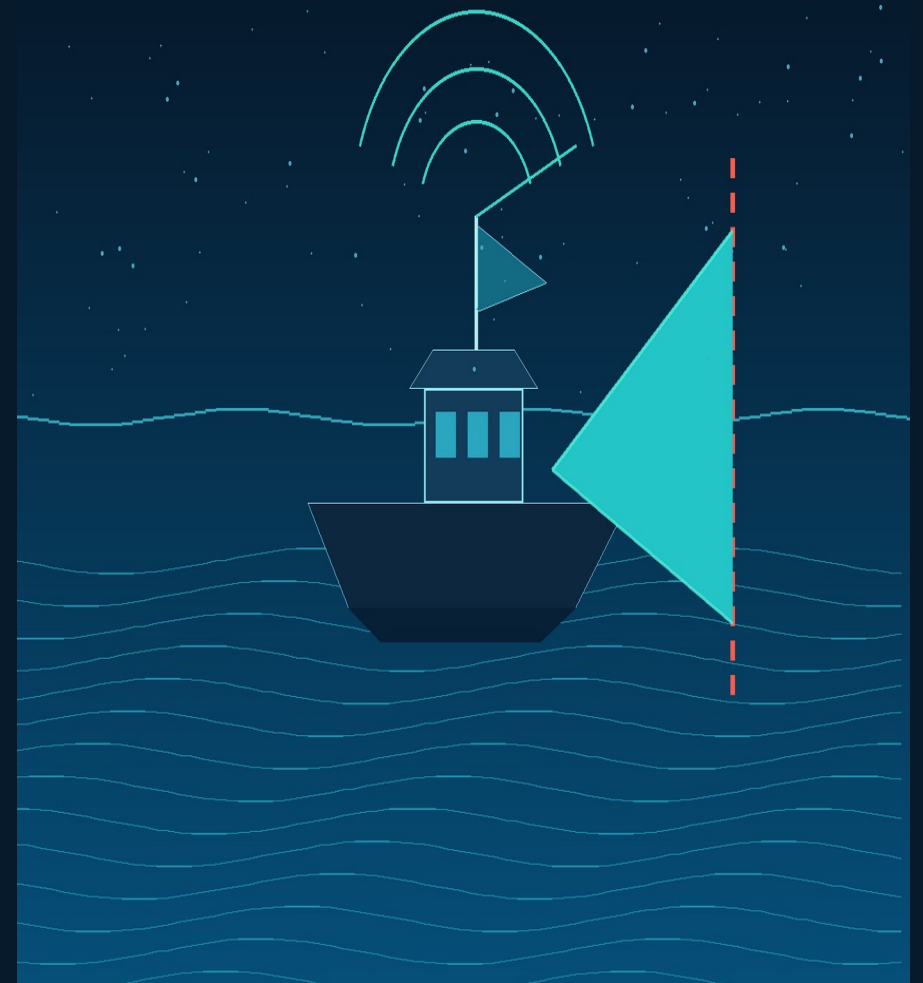


SEA SENTRY

Small-Scale Fisherman Boundary Alert
System

M. Aashika • M. Iniya • G. Deepika

Department of Electronics and Communication Engineering



Introduction

Fishing activities often take place far from familiar landmarks. In such conditions, it can be difficult to notice when a boat or fisherman is moving close to a restricted or unsafe area.

Sea Sentry presents a simple alert concept: as an object approaches a predefined distance boundary, the system gives clear visual and audible warnings.

Safety awareness through timely alerts

The prototype demonstrates distance sensing, microcontroller processing, and multi-level warning output for academic learning.

Problem Statement

Recognizing an approaching maritime boundary or danger zone is challenging when visibility is limited and there are no immediate visual reference points.

A simple, affordable warning system is needed to demonstrate how proximity information can be converted into understandable alerts before the safety limit is reached.

Design Need

- Clear warning states
- Low-cost electronics
- Fast response to changing distance
- Easy operation for a prototype demo

Objectives

01 — Detect

Detect approaching objects using a distance sensor.

02 — Define

Define multiple warning zones using distance levels.

03 — Visualize

Provide visual alerts using an RGB LED.

04 — Alert

Provide audible alerts using a buzzer.

05 — Build

Develop a simple and affordable prototype.

06 — Demonstrate

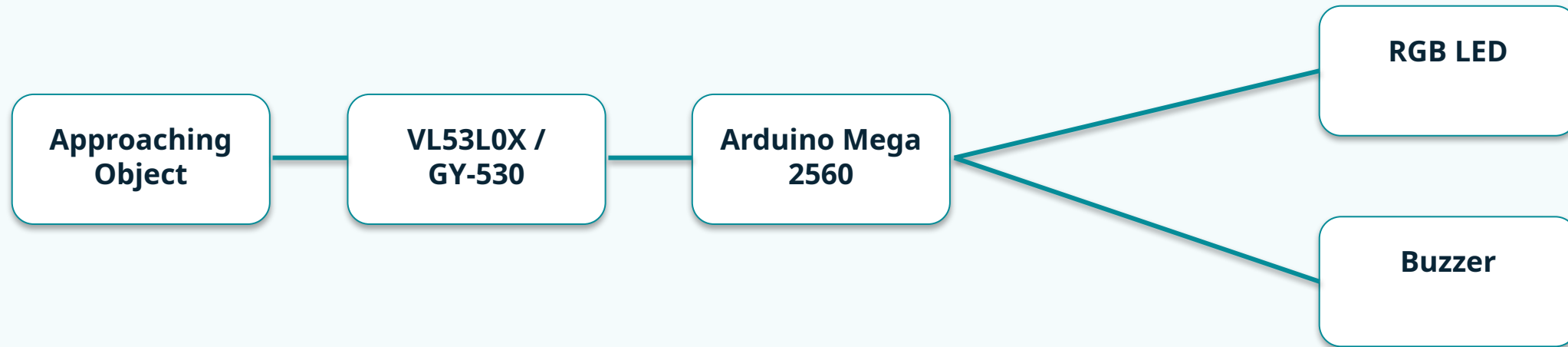
Demonstrate the concept using Arduino Mega 2560.

Proposed System



Object approaches → sensor measures distance → Arduino compares with predefined levels → LED and buzzer provide the appropriate warning.

Functional Block Diagram



The Arduino Mega 2560 receives the sensor reading and controls both warning outputs according to the selected distance zone.

Hardware Components

Component	Purpose in Sea Sentry
Arduino Mega 2560	Main controller that reads distance data and controls warning outputs.
VL53L0X / GY-530	Time-of-flight distance sensor used to measure the approaching object.
RGB LED	Provides green, blue, and red visual warning indications.
Buzzer	Produces different audible warning patterns for each zone.
Connecting wires	Provide electrical connections between all modules.
Breadboard / prototype board	Supports neat assembly and testing of the prototype circuit.

Working Principle

The sensor continuously measures distance. The Arduino assigns the measurement to one of three warning zones.



SAFE ZONE

Object is at a safe distance.

Green LED
No or low warning.



CAUTION ZONE

Object is getting closer.

Blue LED
Moderate buzzer warning.



DANGER ZONE

Object is very close.

Red LED
High or continuous buzzer
warning.

Circuit / Connections

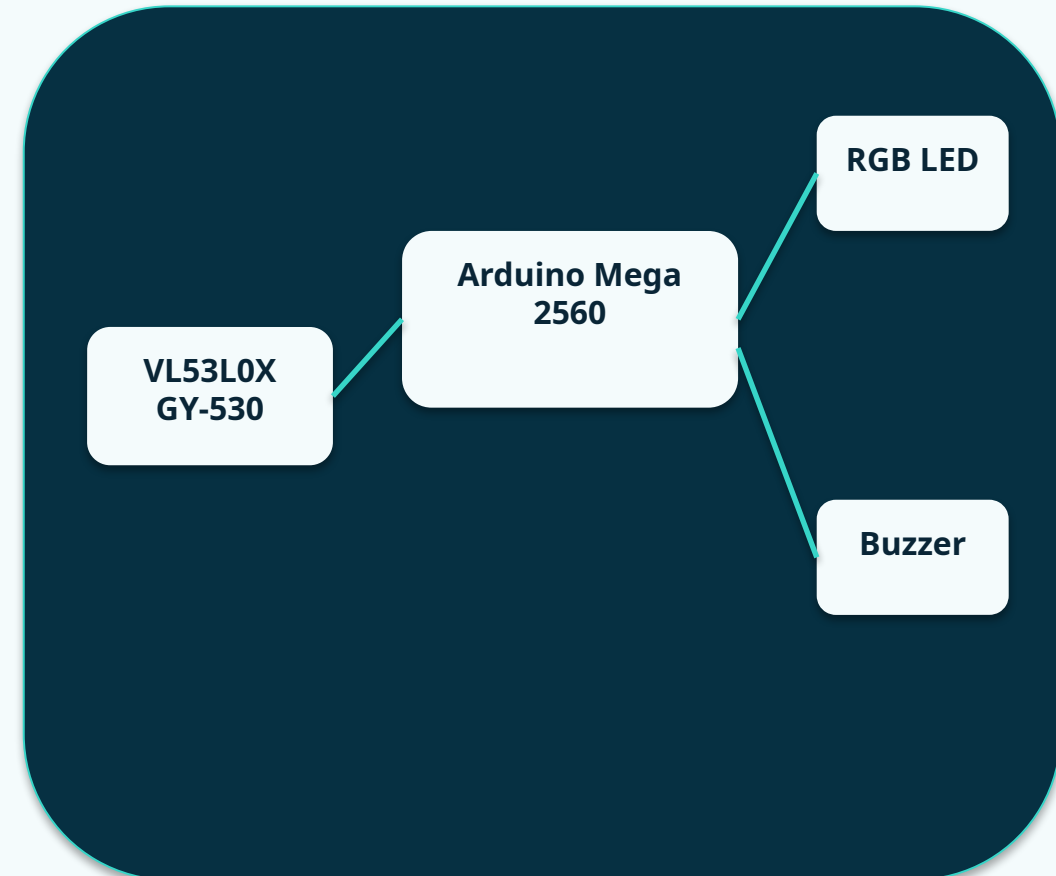
VL53L0X connects to the Arduino Mega 2560 through its I²C interface.

RGB LED pins connect to Arduino output pins for color control.

The buzzer connects to a digital output pin for alert patterns.

All modules share common power and ground connections.

Conceptual connection diagram — pin numbers may be assigned during assembly and programming.



Prototype / Implementation

A working prototype was developed and tested to verify sensor readings and zone-based warning responses.



Prototype Photograph

Insert photo of the assembled Sea Sentry hardware.



Working Prototype Video

Insert QR code or video link during final presentation.



Serial Monitor / Test Output

Insert distance readings and alert-state output.

Advantages and Applications

Advantages

Simple

Low-cost

Easy to understand

Real-time warning

Visual and audible alerts

Suitable for prototype-level demonstration

Applications

Fisherman safety demonstrations

Boundary awareness systems

Small-scale maritime safety prototypes

Educational IoT / embedded-system projects

Conclusion & Future Scope

Sea Sentry demonstrates a distance-based warning system using the Arduino Mega 2560, VL53L0X, RGB LED, and buzzer.

By converting measured distance into green, blue, and red alert levels, the prototype clearly demonstrates the concept of safety-boundary awareness.

This is a prototype concept using a distance sensor; it does not claim real-world GPS maritime boundary detection.

Future Scope

- GPS-based location tracking
- Wireless communication
- Waterproof enclosure
- Improved marine-range sensing
- Integration with a real navigation system